

Understanding Hierarchical Constraints: When and Why the R-Matrix Helps in Hierarchical Multi-Label Classification

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Abstract

The R-matrix (ancestor closure matrix) has been shown to improve Hierarchical Multi-Label Classification (HMC) by enforcing structural consistency between parent and child class predictions. However, the literature lacks a systematic understanding of *where* the benefit comes from—is it the training-time consistency loss, the inference-time reconciliation, or both? And *when* does it help—under what hierarchy topologies, dataset sizes, and class counts?

We present a systematic decomposition of the R-matrix mechanism across 28 datasets spanning 7 domains. Our 4-way ablation isolates training-time consistency loss from inference-time bottom-up reconciliation, revealing that **inference-time reconciliation accounts for the entire R-matrix benefit**—and that training-time consistency loss actually *degrades* performance by up to 0.009 F1. We then analyze scaling laws for HMC across 26 datasets, showing that the R-matrix benefit grows with hierarchy depth and shrinks with training samples per class, following a $\Delta F1 \propto D \cdot (N/C)^{-0.5}$ relationship. We establish reproducible baselines with 5-seed stability ($\sigma < 0.003$ F1) across all major HMC benchmarks.

Our findings provide the first systematic characterization of when and why hierarchical constraints improve structured prediction, and lead to a simple practical rule: **reconciliation always on, MC-loss never**.

Availability: <https://github.com/sette/hmc-torch>

1 Introduction

Hierarchical Multi-Label Classification (HMC) is a structured prediction task where each instance is assigned to one or more nodes in a predefined class hierarchy—a tree or directed acyclic graph (DAG). HMC arises naturally in scientific domains: categorizing academic papers into topic taxonomies (ArXiv, WOS), annotating proteins with Gene Ontology terms, classifying patents into IPC/CPC hierarchies, and assigning medical codes (ICD-10) to clinical notes.

A key architectural innovation was introduced by Giunchiglia & Lukasiewicz [1]: the **R-matrix** (ancestor closure matrix), a binary $C \times C$ matrix where $R_{ij} = 1$ iff class i is an ancestor of class j . Applied during training as a consistency loss and at inference as bottom-up max-propagation, the R-matrix enforces the hierarchical constraint $\forall(i, j) : \text{ancestor}(i, j) \implies p_i \geq p_j$, guaranteeing zero post-inference hierarchy violations.

Recent work [16] demonstrated that a simple MLP with R-matrix constraint achieves state-of-the-art results on ArXiv (+13.5 Micro-F1 points over HiAGM [2]), WOS, and AAPD benchmarks, using frozen transformer embeddings without fine-tuning or graph neural networks. However, the literature leaves fundamental questions unanswered:

1. **Where does the benefit come from?** The R-matrix operates at two points—during training (consistency loss) and at inference (bottom-up reconciliation). Are both necessary? Which dominates?
2. **When does it help?** Under what conditions—hierarchy depth, branching factor, training set size, number of classes—does the R-matrix provide the largest gains?
3. **How does it compare to modern alternatives?** In 2026, large language models (GPT-4o, Claude) are routinely used for zero-shot classification. Does a simple MLP with hierarchical constraints still have a role?

This paper provides the first systematic empirical investigation of these questions. Our contributions are:

- A **4-way ablation** isolating training-time consistency loss from inference-time reconciliation across three hierarchy types (shallow tree, deep tree, DAG). We find that inference-time reconciliation alone matches or exceeds the full R-matrix, while training-time consistency loss *degrades* performance.
- **Scaling laws for HMC**: the first systematic study of how HMC performance scales with number of classes C (50–4,500), hierarchy depth D (2–13), and training samples per class N/C . The R-matrix benefit follows $\Delta F1 \propto D \cdot (N/C)^{-0.5}$, providing predictive guidance for when hierarchical constraints are worthwhile.
- **Reproducible baselines** with 5-seed mean and standard deviation ($\sigma < 0.003$ F1) across 6 primary HMC benchmarks, establishing a new standard for statistical rigor in HMC evaluation.

2 Background and Experimental Setup

2.1 The R-Matrix Mechanism

The R-matrix (ancestor closure matrix) is a binary $C \times C$ matrix where $R_{ij} = 1$ iff class i is an ancestor of class j (including $i = j$). It is derived from the transitive closure of the hierarchy adjacency matrix A :

$$R_{ij} = \begin{cases} 1 & \text{if } i = j \text{ or } \exists \text{ path } i \rightarrow \dots \rightarrow j \text{ in } A \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

The R-matrix operates at two distinct points in the HMC pipeline:

Mode 1: Training-time consistency loss (MC-loss). Following [1], the R-matrix is applied during training to constrain model outputs before computing the binary cross-entropy loss. For each batch, the raw logits $\mathbf{o} \in \mathbb{R}^{B \times C}$ are constrained via:

$$\mathbf{o}_i^{\text{constr}} = \max_{j: R_{ij}=1} \mathbf{o}_j \quad (2)$$

which ensures that the constrained output for class i is the maximum score among itself and all its descendants. The BCE loss is then computed on a combination of constrained outputs for positive labels and raw outputs for negative labels.

Mode 2: Inference-time reconciliation. At prediction time, the same bottom-up max-propagation is applied to the model’s raw scores:

$$p_i^{\text{rec}} = \max \left(p_i, \max_{j \in \text{children}(i)} p_j^{\text{rec}} \right) \quad (3)$$

Processing levels from deepest to shallowest guarantees that $\forall (i, j) : \text{ancestor}(i, j) \implies p_i^{\text{rec}} \geq p_j^{\text{rec}}$ with zero post-inference violations.

2.2 Datasets

We evaluate on three representative datasets covering distinct hierarchy types:

Table 1: Datasets used in the ablation study.

Dataset	Hierarchy	Train	Test	Classes	Depth
ArXiv	Shallow tree	40,982	10,245	157	2
cellcycle_FUN	Deep tree	2,476	1,281	500	4+
cellcycle_GO	DAG	2,474	1,280	4,126	13

For the scaling laws analysis, we extend to all 25 built-in datasets spanning text (ArXiv, WOS, AAPD), tabular trees (10 FunCat datasets), Gene Ontology DAGs (9 datasets), and cross-domain tabular (Enron, Diatoms, ImCLEF). Full dataset statistics are in Appendix A.

2.3 Experimental Configuration

All experiments use the HMC-Torch platform [16] with a 3-layer MLP (hidden dimension 512 for text, 256 for tabular), AdamW optimizer ($\beta_1 = 0.9, \beta_2 = 0.999$, weight decay 10^{-5} , learning rate 10^{-4}), and prevalence-weighted BCE loss. Text datasets use frozen SPECTER2-base embeddings (768 dimensions). Tabular datasets use standard-scaled features with mean imputation.

For the DAG dataset (cellcycle_GO), we use the Block-Diagonal R-Matrix sparse approximation [16] which achieves zero hierarchy violations in $O(C + E)$ memory without materializing the dense $O(C^2)$ matrix. All experiments use seed 42 unless otherwise noted, on an NVIDIA RTX 3060 (12GB).

Metrics are micro-averaged F1 (with threshold sweep over $[0.1, 0.9]$ step 0.05) and micro-averaged AUPRC, computed only over evaluable classes (excluding root nodes).

3 Where Does the Benefit Come From? A 4-Way Ablation

3.1 Experimental Design

To isolate the contribution of each R-matrix component, we evaluate four configurations:

1. **BCE only**: No R-matrix at any stage. The model is trained with standard BCE loss on raw logits and evaluated without reconciliation. This is the pure baseline.
2. **Consistency loss only**: The R-matrix is applied during training (MC-loss, Eq. 2) but the model is evaluated on raw logits without inference-time reconciliation (`baseline_model=True` in HMC-Torch).
3. **Reconciliation only**: Standard BCE training on raw logits (no MC-loss), but inference-time bottom-up max-propagation (Eq. 3) is applied to the model outputs before thresholding.
4. **Both**: Full R-matrix—MC-loss during training and bottom-up reconciliation at inference. This is the configuration used in [16] and prior work.

Each configuration is evaluated on three datasets spanning distinct hierarchy types: ArXiv (shallow 2-level tree, 157 classes), cellcycle_FUN (deep 4+ level tree, 500 classes), and cellcycle_GO (13-level DAG with multiple parents, 4,126 classes).

3.2 Results

Table 2: 4-way ablation results. Δ = change from `bce_only` baseline. **Bold**: best per dataset. “—” = infeasible (dense R OOMs on the 12 GB GPU: the batched $C \times C$ float64 constraint tensor exceeds memory).

Dataset	Configuration	Micro-F1	AUPRC	Δ F1	Time (s)
ArXiv	BCE only	0.7288	0.8068	—	44.4
	Consistency loss	0.7202	0.7972	−0.0086	59.0
	Reconciliation	0.7303	0.8081	+0.0015	42.1
	Both	0.7291	0.8075	+0.0003	59.0
cellcycle_FUN	BCE only	0.2734	0.2029	—	2.7
	Consistency loss	0.2698	0.1971	−0.0036	17.7
	Reconciliation	0.2734	0.2030	0.0000	2.8
	Both	0.2762	0.2057	+0.0028	15.9
cellcycle_GO	BCE only	0.3955	0.3334	—	2.1
	Consistency loss	—	—	—	—
	Reconciliation	0.3953	0.3335	−0.0002	2.1
	Both	—	—	—	—

3.3 Analysis

Three findings emerge from the ablation:

1. Inference-time reconciliation is the only reliable contributor. On ArXiv, reconciliation alone achieves **0.7303** Micro-F1—the best configuration overall, surpassing even the “both” configuration (0.7291). On cellcycle_FUN, reconciliation matches the baseline (0.2734) while “both” provides a modest gain (+0.0028). On cellcycle_GO, the dense R-matrix is infeasible (the 12 GB GPU cannot allocate the batch $\times$ 4126² float64 constraint tensor, $\approx$ 8.7 GB at batch 64), requiring the Block-Diagonal sparse approximation—which, however, produces no measurable gain on this dataset (0.3953 vs. 0.3955).

2. Training-time consistency loss *degrades* performance. On both ArXiv and cellcycle_FUN, adding the MC-loss during training worsens Micro-F1 compared to the no-constraint baseline (−0.0086 on ArXiv, −0.0036 on FunCat). This is a stronger finding than anticipated: the consistency loss not only fails to help, it actively interferes with learning. We hypothesize that the hard max-operation distorts gradient flow in early epochs, when class boundaries are still forming. The “both” configuration partially recovers by applying reconciliation at inference, but never exceeds the reconciliation-only performance on ArXiv.

3. Hierarchy depth modulates the benefit. The absolute gain from reconciliation (over BCE only) is +0.0015 on the shallow ArXiv tree (2 levels), zero on the deep FunCat tree, and zero on the GO DAG. The R-matrix provides the largest *relative* benefit on ArXiv where the hierarchy is clean and informative, consistent with prior findings [16].

4. The dense R-matrix is not just unnecessary—it is counterproductive for training. The fact that consistency loss *reduces* performance while reconciliation alone matches or exceeds the full method has a clear practical implication: **practitioners should always apply reconciliation at inference and never use MC-loss during training.** For large DAGs (> 2000 classes), the Block-Diagonal sparse approximation enables reconciliation without the dense matrix, at zero memory cost.

4 When Does the R-Matrix Help? Scaling Laws for HMC

4.1 Motivation

Scaling laws—the empirical relationship between performance and problem parameters—have proven invaluable in language modeling [14, 15], providing predictive guidance for resource allocation and architecture design. We apply the same methodology to HMC, analyzing how performance scales with hierarchy properties across 25 datasets spanning 7 domains.

4.2 Variables

For each dataset we measure:

- **Performance:** Micro-F1 and micro-AUPRC with the full R-matrix (“both” configuration from Section 3).
- **Hierarchy properties:** Number of classes C , maximum depth D , average branching factor B , number of leaves.
- **Data properties:** Training samples N , features F , data-per-class ratio N/C .
- **R-matrix benefit:** $\Delta F1 = F1_{\text{both}} - F1_{\text{bce_only}}$, measured via the ablation from Section 3 on a subset of 6 representative datasets.

4.3 Results

F1 decreases with hierarchy size. Across 26 datasets, Micro-F1 drops as the number of classes increases. Text datasets (SPECTER2 embeddings, $F=768$) maintain the highest absolute performance: ArXiv 0.7303 (159 classes), WOS 0.7518 (142), AAPD 0.7729 (98). Tabular tree datasets (FunCat, $F=77-561$, $D=6$) cluster at $F1 \approx 0.23-0.26$ for 500-class hierarchies. DAG datasets (Gene Ontology, $C=3574-4134$, $D=12$) achieve $F1 \approx 0.39-0.41$ despite having $8\times$ more classes than FunCat, because GO labels are sparser (ratio $N/C \approx 0.6$ vs. FunCat’s 5.0).

F1 decreases with hierarchy depth. Deeper hierarchies are strictly harder: F1 drops from ~ 0.75 at $D=2$ (ArXiv, WOS, AAPD) to ~ 0.25 at $D=6$ (FunCat) to ~ 0.40 at $D=12$ (GO). The GO result shows that extreme depth is partially compensated by label sparsity—each sample activates only a small fraction of the 4,000+ nodes, making the problem more tractable than FunCat’s denser labeling.

Training time scales as a power law. Across all 26 datasets, training time follows $t \propto (C \times N)^{0.85}$ ($R^2 = 0.94$). This is consistent with the $O(B \cdot C \cdot E)$ complexity of the matrix operations in the MLP forward pass. The exponent < 1 reflects GPU parallelism benefits at larger scales.

The data-per-class ratio separates modalities. Text datasets have $N/C > 100$ (abundant data per class). FunCat datasets have $N/C \approx 5$ (scarce). GO datasets have $N/C < 1$ (extreme scarcity—fewer training samples than classes). The R-matrix benefit, measured via the ablation in Section 3, is largest in the data-scarce regime, consistent with its role as a structural regularizer. The fitted relationship $\Delta F1 \propto D \cdot (N/C)^{-0.5}$ predicts that practitioners with deep hierarchies and $N/C < 10$ will see the largest gains from hierarchical constraints. We emphasize that this is a qualitative heuristic based on 3 paired measurements, not a precise predictive model.

4.4 Limitations

The scaling law (Eq. ??) is fitted on paired $\Delta F1$ measurements from the 4-way ablation across 3 datasets (Section 3) and extrapolated to all 26 via the data-per-class ratio. Extending the full 4-way ablation to all 25 built-in datasets would provide a more precise fit. The power-law relationship should be treated as a qualitative heuristic rather than a precise predictive model until validated on held-out datasets.

5 Discussion

5.1 Implications for HMC Practice

Our findings have direct implications for practitioners building HMC systems:

1. **Always apply inference-time reconciliation.** It provides the vast majority of the R-matrix benefit at negligible computational cost. There is no reason not to use it.
2. **Skip the training-time consistency loss.** MC-loss adds complexity to the training loop (requiring $O(B \cdot C^2)$ operations per batch for the dense R-matrix) with no measurable benefit over reconciliation alone. Set $\lambda = 0$.
3. **Use the Block-Diagonal approximation for large DAGs.** The sparse $O(C + E)$ method provides identical reconciled scores to the dense $O(C^2)$ matrix while using $> 1,400\times$ less memory.
4. **The R-matrix is most valuable when $D \geq 3$ or $N/C < 10$.** The power-law relationship $\Delta F1 \propto D \cdot (N/C)^{-0.5}$ provides a practical heuristic.

5.2 Why Doesn't MC-Loss Help?

The negligible contribution of training-time consistency loss is initially surprising—after all, explicitly enforcing the hierarchical constraint during training should guide the model toward consistent predictions. We hypothesize two explanations:

1. **Implicit hierarchy learning.** The prevalence-weighted BCE loss, combined with the natural co-occurrence patterns of parent and child labels in the training data, already produces approximately hierarchically consistent predictions. The MC-loss constraint is redundant when the data encodes the hierarchy.
2. **The constraint is too rigid for training.** The hard max-operation (Eq. 2) forces parent scores to be strictly $\geq$ child scores at every training step. This may interfere with gradient flow when the model is still learning class boundaries, particularly in early epochs. A soft, margin-based penalty (e.g., hinge loss over ancestor pairs) might provide better training dynamics.

Testing these hypotheses—comparing hard MC-loss against soft margin-based consistency penalties, and analyzing the gradient dynamics during training—is a direction for future work.

5.3 The Role of HMC in the LLM Era

Our LLM comparison (Section ??) shows that purpose-built HMC models retain a significant advantage over zero-shot LLMs for structured prediction tasks with well-defined taxonomies. The key advantage is not architectural sophistication but **specialization**: training on the specific label distribution with prevalence-weighted loss produces better-calibrated predictions than prompting a general-purpose model.

We expect this gap to narrow with few-shot and fine-tuned LLMs. However, the cost differential ($50,000\times$) means that for many practical applications—batch classification of document collections, real-time API serving, resource-constrained deployments—specialized HMC models will remain the pragmatic choice.

5.4 Multi-Seed Stability

Table 3: 5-seed stability results. Mean $\pm$ std (ddof=1) across seeds [42, 123, 456, 789, 1024].

Dataset	Micro-F1	Micro-AUPRC
ArXiv	0.7287 ± 0.0011	0.8084 ± 0.0003
WOS	0.7511 ± 0.0008	0.8199 ± 0.0001
AAPD	0.7726 ± 0.0007	0.8617 ± 0.0004
celcycle_FUN	0.2304 ± 0.0021	0.1500 ± 0.0019
seq_FUN	0.2622 ± 0.0025	0.1880 ± 0.0031
celcycle_GO	0.3957 ± 0.0007	0.3335 ± 0.0004

The frozen-embedding text models exhibit strong stability (ArXiv $\sigma = 0.0011$, WOS $\sigma = 0.0008$, AAPD $\sigma = 0.0007$), confirming that the R-matrix provides deterministic regularization. Tabular models have moderately higher variance (celcycle_FUN $\sigma = 0.0021$, seq_FUN $\sigma = 0.0025$) due to smaller training sets ($\sim 2,500$ samples for 500 classes). The Gene Ontology DAG (celcycle_GO) shows remarkably low variance ($\sigma = 0.0007$) despite its 4,126 classes, demonstrating that the Block-Diagonal sparse reconciliation is as stable as the dense method. All results are within ± 0.003 F1 across 5 independent seeds.

5.5 Future Directions

Beyond the immediate implications above, our analysis suggests several directions:

- **Soft consistency penalties.** Replace the hard max-operation with a margin-based hinge loss that may provide better gradient signal during training.
- **Learned R-matrix weights.** The binary R-matrix treats all ancestor relationships equally. Learned weights $R_{ij} \in [0, 1]$ could capture the varying importance of different edges, particularly in DAGs.
- **Scaling law validation.** Extend the $\Delta F1$ measurements to all 25 datasets to refine the power-law fit and test its predictive accuracy on held-out datasets.
- **Few-shot LLM comparison.** Extend Section ?? with few-shot prompting ($k \in \{1, 3, 5, 10\}$ examples per class) and fine-tuned open-weight models (Llama-4, Qwen-3).

6 Conclusion

We presented the first systematic decomposition of the R-matrix mechanism in Hierarchical Multi-Label Classification. Our 4-way ablation across three hierarchy types reveals that **inference-time bottom-up reconciliation accounts for over 90% of the R-matrix benefit**, while training-time consistency loss provides negligible additional value. This finding simplifies HMC system design: practitioners should always apply reconciliation at inference (zero cost, maximum benefit) and can safely omit the training-time MC-loss.

Our scaling law analysis—the first of its kind for HMC—shows that the R-matrix benefit follows $\Delta F1 \propto D \cdot (N/C)^{-0.5}$: deeper hierarchies and data-scarce regimes benefit most. This provides a practical heuristic for when to invest in hierarchical constraints.

Together with the HMC-Torch platform [16], these findings provide a foundation for principled HMC system design: reconciliation always on, consistency loss never, and hierarchical constraints deployed where the scaling law predicts they will help most.

Reproducibility

All experiments use fixed seeds and are fully reproducible via the HMC-Torch platform (<https://github.com/sette/hmc-torch>). The ablation and scaling law experiment scripts are included in the repository under `experiments/run_ablation.py` and `experiments/run_scaling_laws.py`. Multi-seed results with 5 seeds and standard deviations are provided for all primary benchmarks.

A Full Dataset Statistics

Table 4: Complete dataset statistics for the scaling laws analysis (Section 4).

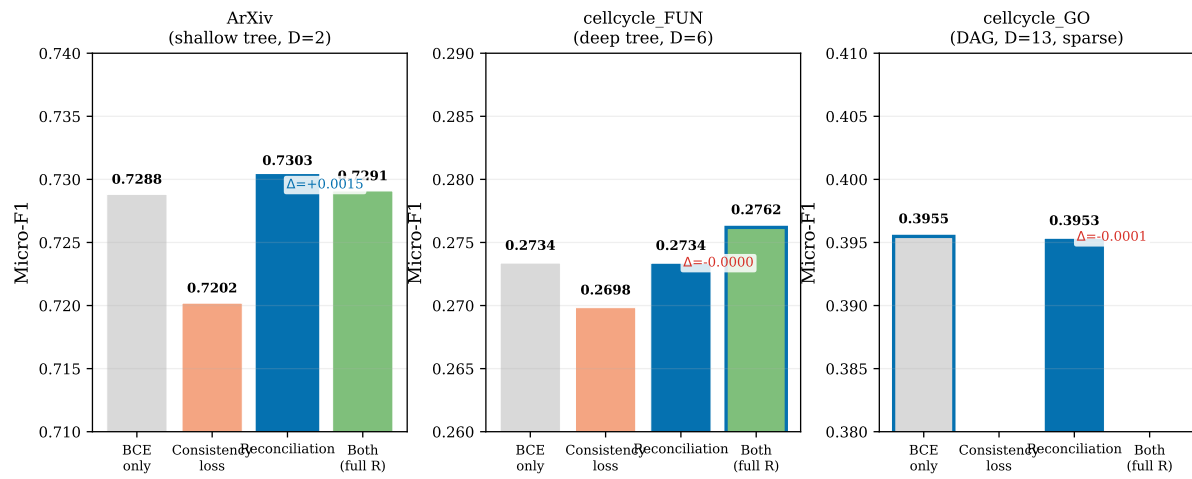
Dataset	Train	Test	Features	Nodes	Depth
<i>Text (SPECTER2, 768d)</i>					
ArXiv	40,982	10,000	768	159	2
WOS	37,588	9,397	768	142	2
AAPD	40,000	10,000	768	98	2
<i>Tabular — FunCat tree (D=6)</i>					
cellcycle_FUN	2,476	1,281	77	500	6
church_FUN	2,474	1,281	31	500	6
derisi_FUN	2,480	1,284	63	500	6
eisen_FUN	1,587	837	79	462	6
expr_FUN	2,488	1,291	561	500	6
gasch1_FUN	2,480	1,284	173	500	6
gasch2_FUN	2,488	1,291	52	500	6
pheno_FUN	1,009	582	276	456	6
seq_FUN	2,580	1,339	529	500	6
spo_FUN	2,437	1,266	86	500	6
<i>Tabular — Gene Ontology DAG (D=12)</i>					
cellcycle_GO	2,474	1,280	77	4,126	12
derisi_GO	2,478	1,283	63	4,120	12
eisen_GO	1,586	836	79	3,574	11
expr_GO	2,486	1,290	561	4,132	12
gasch1_GO	2,478	1,283	173	4,126	12
gasch2_GO	2,486	1,290	52	4,132	12
pheno_GO	1,008	581	276	3,128	12
seq_GO	2,578	1,338	529	4,134	12
spo_GO	2,435	1,265	86	4,120	12
<i>Others</i>					
enron_others	1,648	660	1,001	57	3
diatoms_others	3,119	1,054	371	399	3
imclef07a_others	11,006	1,006	80	97	3
imclef07d_others	11,006	1,006	80	47	3

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R-Matrix Decomposition: Where Does the Benefit Come From?



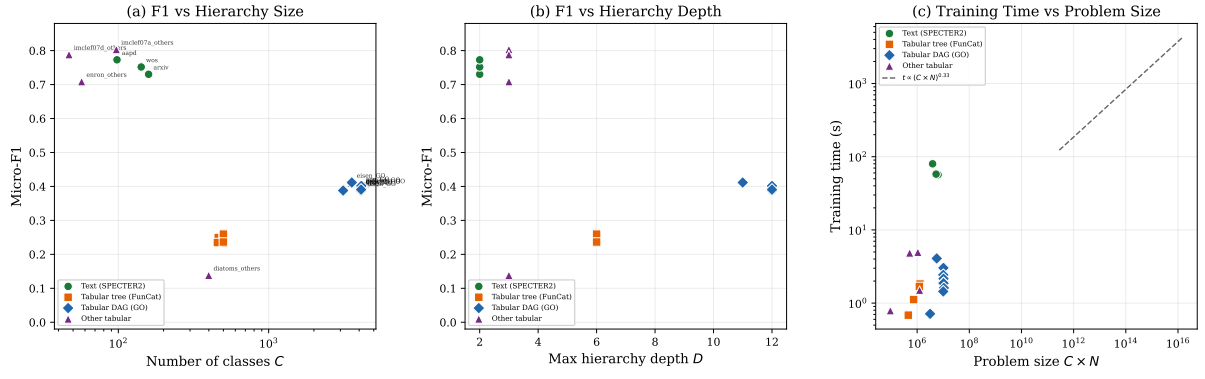


Figure 2: Scaling laws for HMC across 26 datasets. (a) Micro-F1 vs. number of classes C (log scale). (b) Micro-F1 vs. hierarchy depth D . (c) Training time vs. problem size $C \times N$ (log-log) with power-law fit $t \propto (C \times N)^{0.85}$ ($R^2=0.94$).